

Flames with plasmas

Deanna A. Lacoste^{a,b}

^a *Mechanical Engineering Program, Physical Science and Engineering Division, King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Saudi Arabia*

^b *Clean Combustion Research Center, King Abdullah University of Science and Technology, Saudi Arabia*

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Abstract

Non-equilibrium plasma discharges can create a favorable environment for combustion processes, at low energy cost. “How favorable?” and “How low?” are two open questions that can be answered only by gaining a deep understanding of the coupling between flames and plasmas. This paper reviews recent progress in plasma-assisted combustion with a special focus on flame dynamics. First, different non-equilibrium plasma discharges are presented and their effects on flame properties are discussed. Then, an additional level of complexity is introduced by considering the effect of such plasmas on the coupling between flames and acoustic waves. While initial results suggested that the best plasma actuation would combine thermal and chemical effects, a shift towards favoring a chemical impact is emerging. Reasons for this shift are explained and future research directions are finally proposed.

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1. Introduction

A flame can be defined as a thin exothermic reaction front [1]. Its properties depend on the chemical composition of the medium, the thermodynamic conditions, and the flow. By changing any of these, it is possible to modify the combustion rate by several orders of magnitude. In extreme cases, the flame can be extinguished, for example by quenching, blowout, or when the lean flammability limit is reached, or it can turn into another regime of combustion, such as a detonation [2] or MILD combustion [3]. In between, changing the

chemistry, the temperature, or the transport properties will affect the flame burning rate, structure, and temperature. This is the concept upon which plasma-assisted combustion (PAC) has been developed.

When an electric field is applied to a gas, the charged particles are accelerated. If the electric field is strong enough and last for a sufficiently long time, by collision, the electrons will excite, dissociate, and ionize some of the gas neutral particles, generating more electrons that will be accelerated by the electric field, etc. Without going into the details of the physics involved, this process of electric field-induced gas ionization is usually called a breakdown [4]. If after the breakdown a flow of electrons (electric current) is established, then the

E-mail address: deanna.lacoste@kaust.edu.sa

entire process is called an electric discharge or a discharge. At ambient conditions, the density of electrons in air is typically of 10^3 cm^{-3} . After breakdown, the electron density can reach any value between 10^9 and 10^{19} cm^{-3} . For these densities of electrons, one can refer to a plasma of air.

A plasma can be defined as a quasineutral gas of charged and neutral particles which exhibits collective behavior [5]. A system is quasineutral if its characteristic dimension is much larger than the Debye length, λ_D , which can be expressed as:

$$\lambda_D = 69(T_e/n_e)^{1/2} \text{ (m)}, \quad (1)$$

where T_e is the electron temperature in K and n_e is the electron density in m^{-3} . In this definition, “collective behavior” means that the motions of particles depend not only on local conditions (by collisions) but on the state of the plasma in remote regions (by electric or magnetic fields). Therefore, it depends on the electron concentration and energy. With this definition, ambient air, which is a weakly ionized gas, is not a plasma, while an hydrocarbon-air premixed flame at atmospheric pressure, with a typical electron density $n_e = 10^{10} \text{ cm}^{-3}$ [6] and an electron temperature $T_e = 2200 \text{ K}$, is a plasma [7]. Thus, flames can be directly affected by electric fields. For example, by applying an AC electric field to a flame, it is possible to change its shape by ionic wind [8,9] or to reduce the soot formation [10]. Study of the electrical properties of flames, as well as the flame response to external electric fields are interesting parts of the PAC research. Nevertheless, for concision purpose, only interactions between flames and plasmas produced by electric discharges will be discussed in this paper.

Electric discharges turn gas into plasma [4]. Depending on the conditions, this plasma can be in thermal equilibrium (all particles follow Boltzmann distributions at the same temperature), or in thermal non-equilibrium (different particles have different temperatures). Starting from a stoichiometric mixture of air and fuel at standard conditions, the typical temperature of a thermal plasma is 6000–10,000 K. This is three to five times higher than the temperature of a flame burning the same mixture. In addition, the electrical energy necessary to sustain such a plasma is high and the amount of thermal nitric oxide, NO, produced is also high. For these reasons, thermal plasmas have strong limitations for applications in combustion, even though they have been used for more than a century as ignition sources in engines [11]. Comparisons between conventional ignition sources and ignition systems based on non-equilibrium plasmas will be discussed in Section 3.1.

Non-equilibrium plasmas, also called non-thermal plasmas, produced by electric discharges have usually high-temperature electrons (e.g., $T_e = 50,000 \text{ K}$), while the neutral particles (molecules and atoms) are at a much lower temperature, typically between 300 and 4000 K. At elevated pres-

ures (atmospheric and above), due to the fast rate of collisions between particles, thermal equilibrium is usually reached in a very short time. Therefore, the generation of non-equilibrium plasmas requires specific devices. Some of them are presented in Section 2. For applications in combustion, these plasmas are very interesting because they can create reactive chemical species and heat the gas in a controlled manner. Depending on the targeted effect, non-equilibrium plasmas can be tuned to promote the production of different chemical species coupled or not with gas heating.

Plasmas can affect a flame by three main ways: (i) by increasing the gas temperature, (ii) by changing the chemical composition, and (iii) by changing the flow. These effects are generally coupled but, depending on the electric field applied, it is possible to promote one or the other. For thermal plasmas, heating of the gas is the dominant effect but, as a consequence of a strong and fast increase in temperature, turbulence as well as acoustic waves can be generated [12,13]. With non-equilibrium plasmas, the chemical effect can be promoted. Examples of chemical species directly produced by electron impact in hydrocarbon-air discharges are atomic oxygen, O, atomic hydrogen, H, excited oxygen, O_2^* , excited nitrogen, N_2^* , etc. [14]. The way to control the production of these species will be introduced in Section 2. These species are extremely reactive and they can accelerate the combustion processes. For a stronger acceleration, the chemical effect can be coupled with a thermal effect, and/or an increase in mixing and transport, e.g., by ionic wind, turbulence, or shock waves.

Flames with plasmas have been studied with different objectives in mind: from improving ignition to anchoring turbulent flames, from increasing burning rates to extending lean flammability limits. These studies have in common a focus on the plasma efficiency. The plasma energy or power should be as low as possible compared to the expected gain. For example, in a conventional spark ignition engine for road transportation, the energy of a spark discharge is of the order of 100 mJ, which corresponds to roughly $10^{-4}\%$ of the thermal energy released by combustion of the fuel in the cylinder. This is negligible and trying to decrease it further is pointless. Therefore, research in plasma-assisted ignition has focused on conditions for which ignition is difficult, e.g., for very lean or highly diluted mixtures. Then, the gain will be in fuel saving thanks to leaner conditions and/or avoidance of misfire ignition. For combustion systems with a permanent flame such as boilers or gas turbine engines, the plasma power should be as low as possible compared to the flame thermal power. If the focus is on extension of the lean blow-off limit, the plasma power should be compared to the power equivalent of the fuel being saved. Recent PAC studies in these fields are discussed in Section 3.

Finally, the coupling of flames and plasmas have been considered as a potential solution to mitigate thermoacoustic instabilities in combustion systems. Thermoacoustic instabilities can happen when the fluctuations of heat release rate from a flame couple with an acoustic mode of the combustion system [15]. These instabilities can cause strong thermal and pressure fluctuations that can threaten the integrity of the engine [16]. Control of thermoacoustic instabilities have been intensively studied, using both passive and active approaches [17,18]. Recently, active approaches using non-equilibrium plasma discharges to control flame dynamics have been successfully attempted [19–27]. Section 4 presents recent findings in this field and propose a discussion to get closer to real engine conditions.

The main objectives of this paper are the following:

- If you are a PAC curious or beginner, to help you understand the important points of this research (Sections 1–3).
- If you already have experience in PAC, to present the current state of the art and some of the remaining open questions (Sections 3 and 4).
- If you are an expert in PAC, to challenge some of your ideas and to encourage you to try to prove me wrong (Sections 3.3, 4.2, and 4.3).

2. Non-equilibrium plasma sources

Different sources can be used to produce non-equilibrium plasmas by electric discharges at atmospheric pressure and above. Some of the most commonly used in PAC studies are presented in this section.

As briefly discussed in the introduction, an electrical discharge can be decomposed in two phases, the breakdown and the subsequent flow of electrons in the ionized channel. If nothing limits the electric current after breakdown, the plasma will rapidly turn into a thermal plasma. To avoid transition to a thermal plasma, there are two simple solutions. The first one is to stop the discharge shortly after the breakdown and the second option is to control the flow of electrons, *i.e.*, the current. Thus, applying short duration electric fields, *i.e.*, typically shorter than 100 ns, or using devices to avoid strong currents are the two main ways of producing non-equilibrium plasmas by electric discharges. For applications in PAC, other parameters must however be considered.

In weakly ionized plasma chemistry, *i.e.*, for $n_e \ll N$, where N is the gas number density, the reduced electric field, E/N , where E is the electric field, and the electron density, n_e , are the two key parameters [29]. Electron temperature and

electron-impact reaction rates both strongly depend on E/N . The reduced electric field controls the modes into which the electron energy is dissipated [30]. For example, Fig. 1 presents the fraction of dissipated energy by electrons during a discharge in air [28]. At low reduced electric field, $E/N < 5$ Td, the main fraction of dissipated energy per electron goes into vibrational excitation of oxygen. For $5 \text{ Td} < E/N < 100 \text{ Td}$ vibrational excitation of nitrogen is the main pathway, while at $E/N > 100 \text{ Td}$, electronic excitation of nitrogen dominates. Thus, the chemistry induced by the discharge can be tuned by controlling the reduced electric field. While the gas density depends on the thermodynamic conditions of the gas, the electric field depends on the applied voltage and the geometry and gap distance of the electrodes.

Finally, the last important parameter to consider for a source of non-equilibrium plasma is the repetition rate. Except for single shot applications such as ignition, PAC relies on long-duration non-equilibrium plasmas. The repetition rate required for a given application depends on the thermodynamic and flow conditions. Indeed, to maintain a sufficient average level of chemical species and/or gas temperature in the discharge, the recombination rates as well as the residence time in the inter-electrode area must be taken into account.

2.1. Dielectric barrier discharges (DBD)

Dielectric barrier discharges (DBD) are produced by applying an electric field, generally AC or pulsed, between two electrodes separated by a gas and one or more dielectric layers [31]. At atmospheric pressure, if the electric field is strong enough, breakdown occurs in the gas, and an ionized channel joining one electrode and the dielectric surface forms. Electrons flowing in the discharge accumulate on the dielectric surface, building up a potential that finally suppress the local electric field. Thus, the discharge stops before the plasma could reach thermodynamic equilibrium. Multiple discharges are produced in parallel, at other locations of the electrodes/dielectric surface, and DBD discharges at elevated pressures are generally inhomogeneous. When the electric field is reversed or stopped, electrons accumulated on the dielectric surface are removed by the reverse electric field or by recombination. In some cases, they may contribute to the breakdown of the next half period. Fig. 2 presents examples of DBD geometries used in PAC as well as an example of measured voltage and current, obtained in a DBD for an AC applied voltage of 7 kV at 4 kHz.

At atmospheric pressure, DBD are a common and cheap way of producing non-equilibrium plasma with a negligible thermal impact. They have been used for more than a century to generate ozone from air. Different types of high-voltage

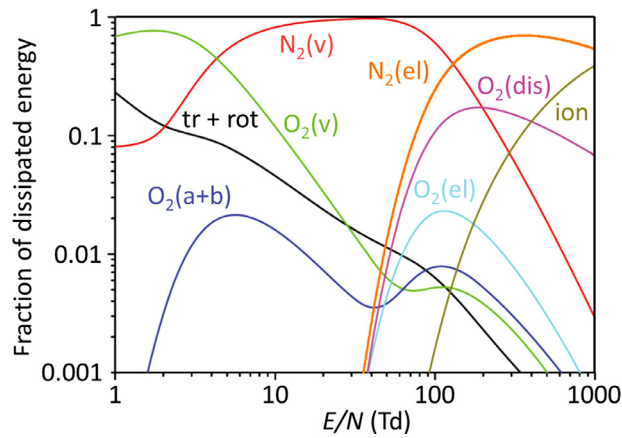


Fig. 1. Fraction of dissipated energy by electrons for a discharge in air as a function of the reduced electric field, E/N . tr + rot, sum of translational and rotational excitation; v, vibrational excitation; el, electronic excitation; ion, ionization; dis, dissociation; a + b, sum of electronic excitation of singlet ‘a’ and ‘b’ states of molecular oxygen. Data from [28].

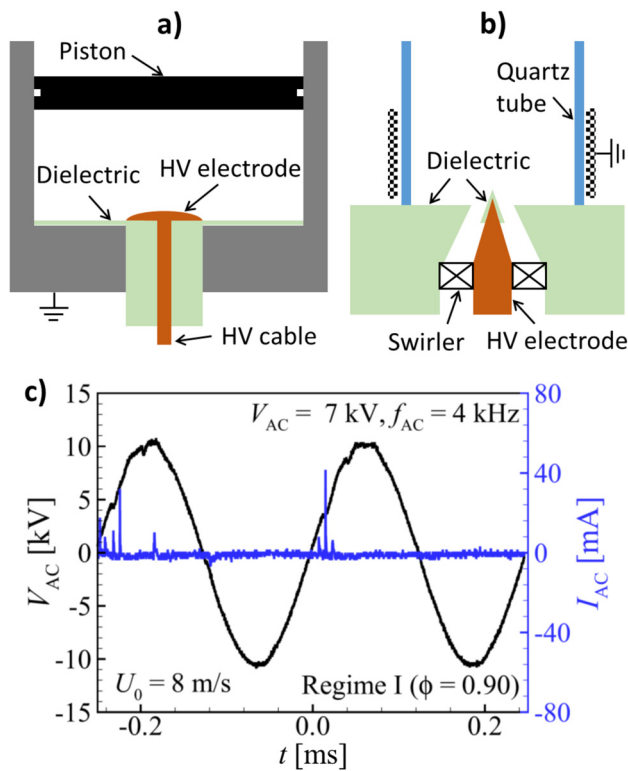


Fig. 2. (a) Example of DBD configuration used for plasma-assisted ignition in [32]. (b) Example of DBD configuration used for swirl flame stabilization in [33]. (c) Example of current and voltage signals measured during swirl flame stabilization, taken from [33].

sources can be utilized, from the most basic AC or pulsed DC systems, to nanosecond repetitively pulsed ones. For applications to combustion, the typical range of frequency used is from 1 to 20 kHz, but lower or higher frequencies can also work. For

DBD produced by AC or DC pulsed high-voltage, in addition to the chemical actuation of the gas, transport effects can also be observed, from the so-called ionic wind [34]. This effect can be suppressed by using nanosecond pulsed high-voltage

sources. In this case, the acronym ns-DBD is sometimes used. Another advantage of ns-DBD is that the reduced electric field at which the discharge occurs can be controlled, while in traditional AC-voltage DBD, discharges can happen at various E/N , therefore the chemical production is less controlled. Fig. 2 c shows an example of AC-voltage DBD where discharges, corresponding to the current peaks, happen at voltages from 2 to 10 kV [33]. These multiple discharges happening during a cycle are one of the interesting features of DBD. Indeed, if large electrodes can be used, the plasma volume produced by the multiple discharges can be relatively large. Finally, the best advantage of DBD is also their main limitation: the thermal effect is minimal. For combustion applications where an increase in gas temperature is desirable, DBD might not be suitable sources of non-equilibrium plasmas.

2.2. Gliding arcs

A gliding arc occurs when a plasma discharge is generated between two or more diverging electrodes placed in a fast gas flow [35]. When a strong enough electric field is applied, breakdown happens at the shortest distance between the electrodes. Rapidly, *e.g.*, within 1 μ s in ambient air, the discharge turns into an arc, *i.e.*, a thermal plasma with a temperature controlled by the maximum power of the power supply (typically in the range 6000–10,000 K). This arc is placed in a flow and it is convected along the diverging electrodes at a speed that is usually comparable to the bulk flow velocity. During this movement, the length of the discharge and hence its dissipated power increase up to the moment when the discharge power reaches the maximum value available from the power supply source. When the distance between the electrodes reaches a critical value, heat losses from the plasma begin to exceed the energy supplied by the source and it is not possible to sustain the plasma in the state of thermodynamic equilibrium. Consequently, the translational temperature of the neutral molecules decreases while the plasma conductivity is maintained by a high value of the electron temperature. This new mode of the gliding arc is non-equilibrium. When the gap between the electrodes becomes too large, there is a decay of the non-equilibrium discharge, followed by a new breakdown at the shortest distance between the electrodes and the cycle repeats. Fig. 3 presents examples of gliding arc electrode configurations used in flame stabilization studies [36,37], along with an example of voltage and current signals measured in a gliding arc in air, powered by an AC voltage at 12.5 kHz [37].

Gliding arcs have been studied since the early 1990's. They operate at elevated pressures and they are easy and cheap to produce with DC or AC high voltage sources. They can dissipate large pow-

ers, *i.e.*, tens of kilowatts for a single pair of electrodes. They produce a mix between equilibrium and non-equilibrium plasmas, combining strong gas heating with transport effects due to gas expansion and shock/acoustic waves, with chemical actuation of the gas. For all these reasons, they are a very interesting source of non-equilibrium plasma for combustion applications. However, they must be set aside for applications where a thermal effect is beneficial.

2.3. Microwave discharges

Discharges produced by high-frequency electromagnetic fields are referred to as microwave discharges [38]. Electrons in the gas are accelerated by the microwave electric field, typically at a frequency of 2.45 GHz, and collide with the neutral particles. At the beginning of the applied field, these collisions are elastic and they result in an increase of the gas temperature. After several collisions, the electrons get enough energy to produce ionization. The gas is then partially ionized and becomes a plasma. In a microwave discharge, only the electron are accelerated by the electric field, as ions are too heavy to follow the oscillating electric field. The microwave energy transmitted to the electrons has to be high enough to initiate the plasma. To concentrate this energy, different methods have been developed.

A resonant cavity [39] can be used to generate a standing wave that is amplified by the waves produced by the microwave source. The resonance frequency of the cavity strongly depends on its geometry and has to match the frequency of the source. While resonant cavity microwave sources are minimizing the losses, this geometrical constraint is one of the limitation for their application in PAC. A second method is to work with a wave guide-to-coaxial line transition [40]. The microwave plasma is produced on top of a nozzle. It has the shape of a converging cone. It is sometimes called a free expanding torch. Several designs of microwave plasma torch are available [38]. In PAC, this technique is interesting because it is electrode-less. A third method consists of a planar microstrip line on a dielectric substrate and a copper ground plate. Surface waves propagate in a gas channel along the interface between the plasma and the dielectric surface [41]. This technique produces low-power microplasmas.

Microwave discharges have been studied since the 1970's. They are well understood but the physics involved is more complicated than for other types of non-equilibrium plasma discharges. In addition, their thermal effect is usually significant, even though pulsed microwave systems can be used to control it [42,43]. On the other hand, they have advantages such as the ability to impact large volumes [44,45], a low cost, or electrode-less configurations [46], that make them interesting sources for PAC.

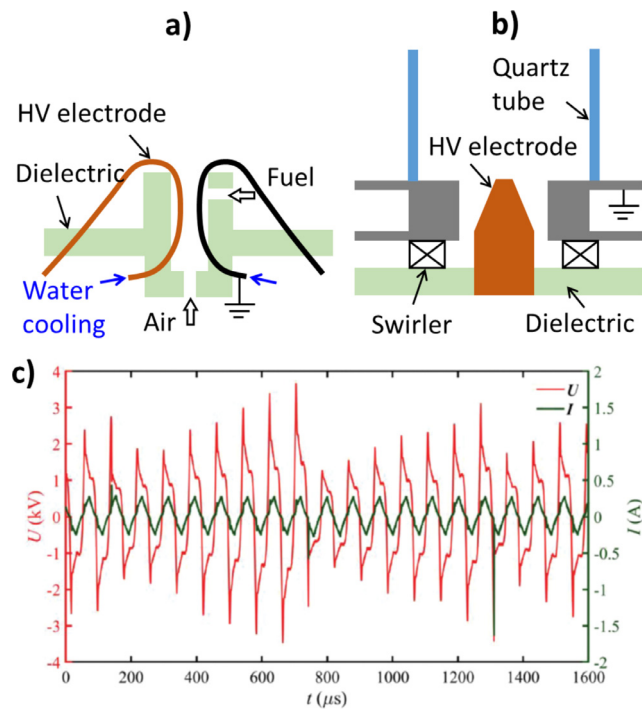


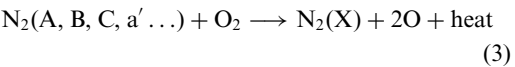
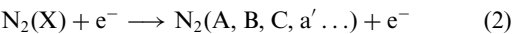
Fig. 3. (a) Example of gliding arc electrode configuration used for jet flame stabilization in [36]. (b) Example of gliding arc electrode configuration used for swirl flame stabilization in [37]. (c) Example of current and voltage signals measured in air for an AC gliding arc at 12.5 kHz, taken from [37].

2.4. Nanosecond repetitively pulsed (NRP) discharges

Nanosecond repetitively pulsed (NRP) discharges are produced by applying nanosecond high-voltage pulses at high repetition rate, typically between 5 and 200 kHz [47]. The voltage pulses have a duration between 2 and 100 ns and a very sharp rising slope, between 1 and 10 ns. These features constrain the breakdown to occur at electric fields much higher than for continuous applied voltages. After breakdown, a strong current controlled by the power supply flows in the ionized channel. The electric field drops and the temperature of the neutral particles increases very rapidly. Nonetheless, the discharge is stopped long before the plasma could reach thermodynamic equilibrium. Afterward, the charged and excited particles undergo a post discharge chemistry. Before the state of equilibrium is reached, *i.e.*, charged particles, excited particles, atoms, etc. remain at relatively high concentration in the interelectrode area, a new pulse is applied. This synergistic effect between pulses is one of the key features of NRP discharges because it allows to maintain a high concentration of active species at low energy cost. Fig. 4 presents an example of a train of 10 NRP discharges in air [48]. During the first five pulses the current is low and the voltage is high, while after the

sixth pulse, high current peaks and low voltage peaks, corresponding to a steady state regime, can be observed. In this example, the building-up process took five pulses before the synergistic effect between pulses reached an almost steady state regime.

NRP discharges have been studied since the beginning of the 2000s. In air, their chemistry is relatively well known [49–51]. During the pulse, the main part of the electron energy goes into (1) dissociation of oxygen, (2) excitation of the electronic states of nitrogen, $N_2(A, B, C, a' \dots)$, and (3) excitation of the vibrational states of nitrogen in its ground electronic state, $N_2(X)$. The distribution between these different pathways depends on the reduced electric field (see Fig. 1). During and after a pulse, electronically excited nitrogen reacts with molecular oxygen very rapidly, *i.e.*, within a few tens of nanoseconds. This process is exothermic, therefore leading to a fast heating of the gas. This model, first proposed by Popov in 2001 [49], and verified experimentally [50], is known as the two-step mechanism of atomic oxygen production and gas heating:



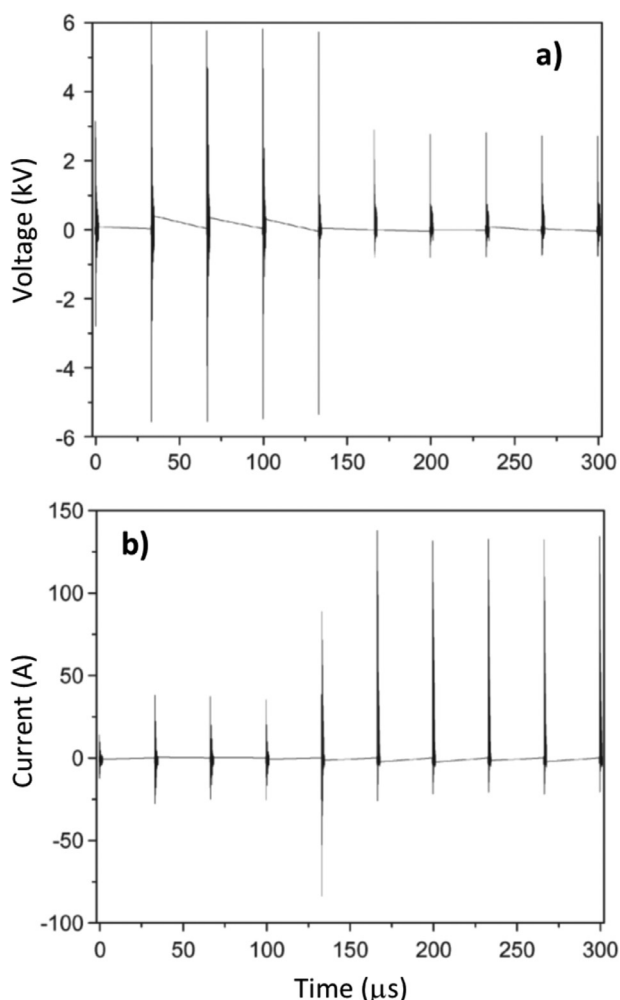


Fig. 4. Examples of voltage and current measured for a train of 10 NRP discharges at 30 kHz in air. Taken from [48].

After a pulse, the vibrational levels of nitrogen molecules eventually reach equilibrium with the translational mode, slowly releasing heat [52]. In parallel, O atoms start to react and, depending on the conditions, produce O_3 , N_2O , NO ... More details on fast gas heating and chemistry of NRP discharges in air and fuel-air mixtures can be found in a recent review paper by Popov and Starikovskaia [51].

For combustion applications, NRP discharges are extremely interesting because by playing with the reduced electric field, pulse duration, and repetition frequency, the electron energy pathway can be controlled. For example, for low applied voltages and relatively low pulse repetition frequencies (5–30 kHz), the glow regime of NRP discharges [53] is promoted, with mainly a chemical effect. When the applied voltage is further increased, the spark regime of NRP discharges [54,55] can be obtained with a strong fast gas heating, inducing gas expan-

sion and shock/acoustic waves [56,57]. Finally, for a fast repetition rate and a low applied voltage, the reduced electric field can be low enough to favor vibrational excitation of nitrogen and a slow gas heating. Other interesting features of NRP discharges are: they are easy to synchronize with optical diagnostics, the electrode geometry can be completely arbitrary, and the discharge power can be as high as a few kilowatts. One of the limitations of this source of non-equilibrium plasma is the cost of the power supply that is usually significantly higher than AC or DC power supplies for DBD and gliding arcs.

2.5. Location of the plasma actuator

Once a non-equilibrium plasma source has been chosen, an important decision is where to install the plasma actuator in the combustion system. There

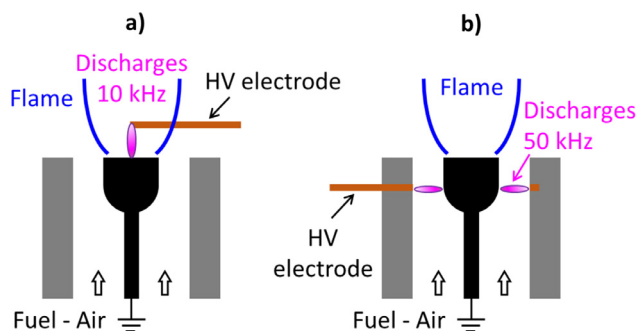


Fig. 5. Examples of location of the non-equilibrium plasma discharges for stabilization of a premixed jet flame over a bluff-body. a) Discharges located in the recirculation zone, repetition rate 10 kHz. b) Discharges located in the jet, repetition rate 50 kHz.

are several possible options and the best one depends on both the application and the system.

For ignition, an obvious guideline is to install the plasma actuator in a zone where the fuel-air mixture is in the flammability range. In addition, the flow properties as well as the presence of walls have to be considered. The ignition kernel should have enough time to develop and should not lose too much heat to a surface or by stretching [58–60]. On the other hand, flow and turbulence can also enhance the mixing and increase the ignition kernel wrinkling, helping its fast development [61,62]. In industrial systems, a good starting point is to choose the same location than the traditional spark ignition system because it usually has been optimized. In laboratory setups or new designs, a good knowledge of the flow field and gas composition is necessary to select the best ignition location [63]. Finally, other constraints such as optical access might also be important to consider [43].

For flame actuation, e.g., increase of the burning velocity, decrease of the lean blow-off limit, control of thermoacoustic instabilities, etc., several strategies have proven to be efficient. From localized discharges in the burned gases [27,64], in the fresh gases [21], or across the flame [65,66], to distributed discharges in the fresh mixture [36,67], in the fuel stream [68] or in the flame front [9], flame actuation by non-equilibrium plasma has been achieved. Here again, a good knowledge of the flow field and gas composition is important but it is not enough. Indeed, it is crucial to understand the flame dynamics or stabilization mechanisms.

For example, consider a premixed jet flame stabilized over a bluff-body. If the purpose of the plasma actuation is to anchor the flame over the bluff-body while decreasing the burning velocity (by changing the temperature or the equivalence ratio), or increasing the flow speed, then, a good strategy is to reinforce the stabilization effect of the bluff-body by applying localized discharges in its recirculation zone, i.e., in the slow burned gases on top of it (Fig. 5a). For example, this location

was chosen in [20,64], with relatively low discharge frequencies (e.g., 10 kHz). Another possible option is to install the plasma actuator on the side of the bluff-body, to enhance the reactivity of the flow of fresh mixture and increase the burning velocity of the flame (Fig. 5b). This strategy requires a faster repetition rate (typically 50 kHz or higher), because the flow velocity is much higher than in the recirculation zone. Thus, the best option would depend on the capabilities of the plasma source.

Only a few studies have compared the effect of the plasma location on the expected result (see for example [69]). More work is required to optimize this parameter for various configurations. Having a systematic approach, maybe coupled with machine learning tools, is necessary to further improve the effectiveness of plasma actuators.

3. Flame enhancement

Flame enhancement by non-equilibrium plasma, also called plasma-assisted combustion (PAC), has been studied for a few decades and showed very interesting results. Several review papers [14,70–72] have summarized both the achievements and the mechanisms of plasma action. In this section, some of the key results on ignition, burning velocity, and lean blow-off and stability limits are summarized. A short section on modeling and simulations is also proposed.

3.1. Ignition

Studies of ignition by non-thermal plasma discharges can be grouped in three large families, from the most fundamentals, using shock tubes and rapid compression machines (RCM), to the most applied using piston engines or scramjets. In between, constant volume combustion chambers (CVCC) and continuous-flow burners have been used to compare various ignition sources.

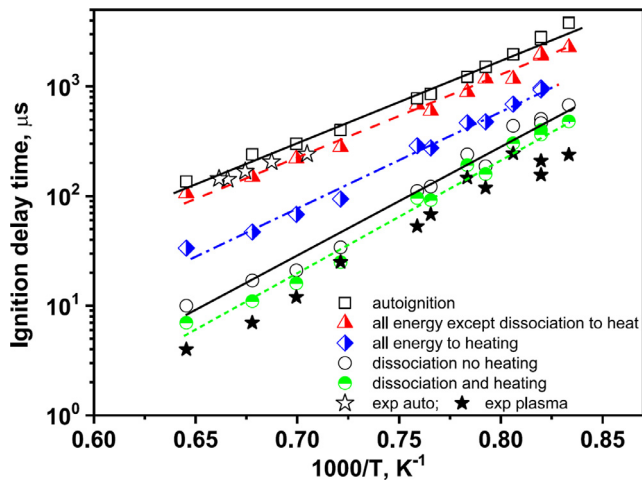


Fig. 6. Measured and calculated ignition delay times in a mixture of $\text{C}_2\text{H}_6:\text{O}_2:\text{Ar}=2:7:81$ as a function of $1000/T$. Shock tube facility equipped with a volumetric nanosecond plasma system. Taken from [73].

To determine the plasma effect on ignition delay time, shock tubes and rapid compression machines have been widely utilized. By comparing the ignition delay times obtained with and without discharges, it is possible to quantify the plasma effect on the combustion chemistry. Coupled with chemical modeling and simulations, these studies helped in understanding the key chemical mechanisms of plasma action on ignition. For example, Kosarev et al. [73] studied in a shock tube the effect of a volumetric nanosecond plasma discharge on the ignition delay time of ethane-oxygen-argon mixtures (2:7:81) at temperatures from 1200 to 1500 K. Fig. 6 presents a comparison between their experimental and modeling results. Two main conclusions can be drawn from this figure. First, over the entire range of temperatures, the use of nanosecond volumetric discharges decreased by more than an order of magnitude the ignition delay time (empty black squares for autoignition delay time and solid black stars for ignition delay time with plasma). Second, they obtained a very good agreement between experimental results and simulations considering dissociation of oxygen coupled with gas heating (solid black stars and green symbols). Shock tubes and RCM are used in PAC research to validate chemical models but also to compare different non-equilibrium plasma strategies. Unfortunately, the number of plasma-assisted ignition studies focusing on ignition delay time is low and only a few fuels and plasma sources have been investigated. More research is needed to validate chemical models of PAC and to better understand the chemical effect of non-equilibrium plasmas on the combustion of fuels with large carbon content.

The second family of experiments used for ignition studies comprises CVCC and continuous-flow burners. These setups allow the investigation

of ignition mechanisms by non-equilibrium plasma discharges including all potential effects (chemical, thermal, and transport). These systems are the most commonly used in plasma-assisted ignition research and a large range of conditions has already been covered. Different initial pressures, flow conditions, fuels, mixture composition, and plasma sources have been considered [13,48,71,72,74]. Typically, plasma sources are compared with conventional ignition systems, which produce thermal plasmas. Lean or diluted flammability limits and propagation speed of the ignition kernel are two common criteria of comparison. Additionally, spectroscopic measurements and visualization can be performed and provide information about the dominant plasma effect. However, a precise understanding of the plasma action is usually not achieved because, in these experiments, it is difficult to decouple the different effects. The vast majority of these studies can be regarded as proof-of-concepts, showing that for a similar energy deposition, non-equilibrium plasmas are more efficient ignition sources than conventional ignition systems producing thermal plasmas. For example, in Xu et al. [75], ignition of lean propane-air mixtures by NRP discharges was compared to a conventional ignition system, in a CVCC at initial pressures up to 10 bar (see Fig. 7). For similar energy deposition (about 55 mJ) and ignition duration (about 3 ms), the flame developed faster in the case of NRP ignition, compared to conventional ignition. These results were attributed to an increase in the flame wrinkling, *i.e.*, to local changes in the flow properties.

Finally, a few studies of plasma-assisted ignition by non-equilibrium plasmas in real engine conditions [76–79], or in scramjets [60,80,81] have been reported. Here, the challenges come from pressures

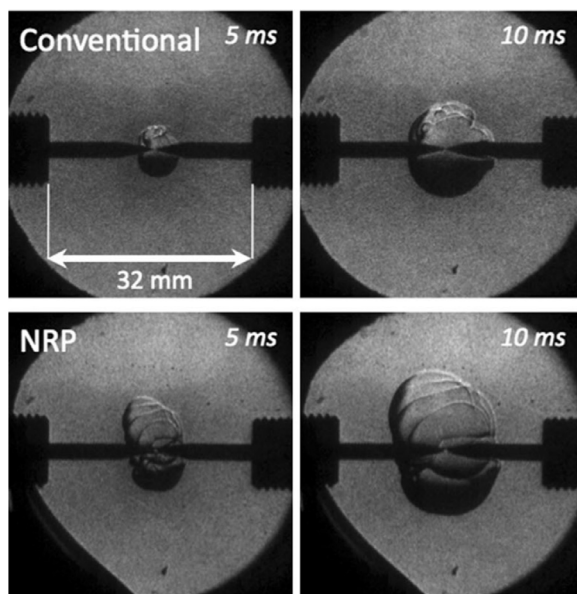


Fig. 7. Schlieren images of lean propane-air flames at 10 bar ignited by a conventional car ignition system (57 mJ; 16 W; 3.5 ms) and 82 NRP discharges at 30 kHz (55 mJ; 22 W; 2.7 ms). Taken from [75].

and turbulence levels higher than in lab scale experimental setups. In addition, implementation of non-equilibrium plasma sources and diagnostics is also more challenging. These papers confirmed that, under engine conditions as well, compared to thermal plasmas, plasma sources combining chemical, thermal, and transport effects could improve ignition.

For ignition of fuel-air mixtures, three effects of non-equilibrium plasma discharges have proven to be important: (i) the decrease of ignition delay time due to the production of active chemical species, (ii) the acceleration of chain branching and chain prolongation reactions due to fast temperature increase, and (iii) the hydrodynamic perturbations and resulting mixing, induced by gas heating. This last effect is not always present and not always beneficial for ignition. For example, in Castela et al. [82], a simplified model of the discharges for ignition of turbulent flows is proposed. This model distinguishes two main electron pathways: (1) the excitation and the subsequent relaxation of the electronic states of nitrogen, leading to a fast gas heating associated with molecular oxygen dissociation within the discharge characteristic time; and (2) the excitation and relaxation of vibrational states of nitrogen, causing a much slower gas heating. In this model, a fraction α of the electron energy goes into dissociation of oxygen, a fraction β goes into fast gas heating, and a fraction γ goes into slow gas heating, with $\alpha + \beta + \gamma = 100\%$ of the energy of the discharge. They showed that if β is too large, a strong gas expansion associated with the generation of shock and/or acoustic waves happens.

This process can remove a part of the energy from the ignition area, reducing the ignition efficiency of the discharges. In addition, for multiple discharges such as in bursts of NRP discharges, the generated flow becomes the dominant process controlling ignition, *i.e.*, favoring or being detrimental to the development of the ignition kernel [13,83,84].

3.2. Burning velocity

While successful plasma-assisted ignition is relatively easy to achieve, enhancing the burning velocity by non-thermal plasma is more challenging. To accelerate the burning velocity of a premixed flame three options are possible. First, by increasing the reactivity and/or the temperature of the gas mixture, the laminar burning velocity can be increased. Second, by changing the flow properties or introducing inhomogeneities (temperature or mixture composition), an increase of the flame surface area can be achieved. Finally, the overall burning velocity of a flame can be increased if the occurrence of flame local extinction (*e.g.*, by stretch or curvature effects), is decreased. Non-equilibrium plasma can be efficiently used for each of these options [85–89]. However, in many configurations at atmospheric pressure and above, the flame volume is much larger than the discharge volume. A local plasma can struggle in accelerating a large flame in such a way that, globally, the burning velocity is accelerated. Anyway, solutions exist and it is sometimes possible to accelerate large flames with localized discharges. Here, two examples are detailed.

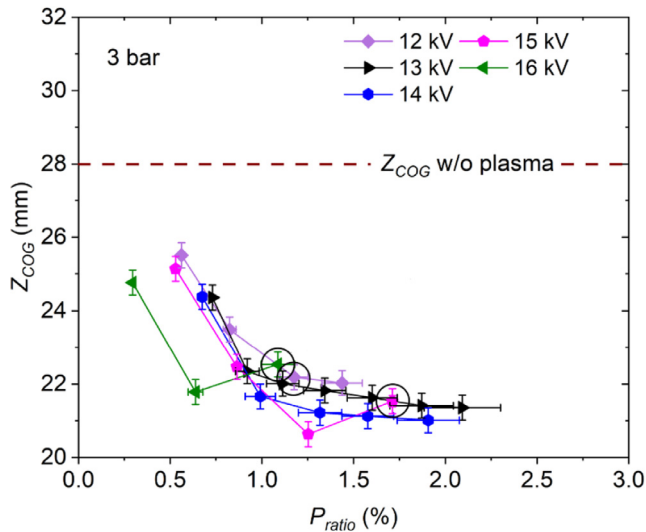


Fig. 8. Distance between the burner plate and the center of gravity of the flame, Z_{COG} , as a function of the plasma to flame power ratio, P_{ratio} , at 3 bar, for different applied voltages. Adapted from [90].

For configurations such as bluff-body stabilized turbulent flames or turbulent swirl flames, non-equilibrium plasma discharges applied close to the stabilization area can decrease the flame height and volume, *i.e.*, decrease the overall burning velocity. For example, Di Sabatino et al. [90], showed that the height of the flame's center of gravity, Z_{COG} (based on imaging of the flame OH^* chemiluminescence, see details in [90]), of a swirl-stabilized methane-air flame could be decreased by up to 25% by applying NRP discharges with a plasma power of about 1.2% of the flame thermal power. Fig. 8 compares the Z_{COG} for the reference flame without plasma (dash line), at 3 bar, with the Z_{COG} of the same flame subjected to NRP discharges, for applied voltages ranging from 12 to 16 kV, and for ratios of plasma power to flame thermal power from 0.25 to 2%. As the bottom part of the flame was not noticeably shifted inside the injection tube, these results illustrate that with NRP discharges, the flame was more compact. In addition, as the flow and gas mixture composition were kept constant, a more compact flame was then the consequence of a faster burning rate. However, reasons for this effect were not discussed.

Another example of burning velocity acceleration by non-equilibrium plasma discharges can be found in Gray and Lacoste [69,91]. In these studies, the objective was to accelerate a turbulent stoichiometric hydrogen-air flame propagating in a tube until the onset of a detonation, *i.e.*, until a deflagration-to-detonation transition (DDT) occurred. Non-equilibrium plasma produced by NRP discharges were used in burst mode (less than 100 discharges) and phased with the arrival of the turbulent flame in the interelectrode area. The flame

could be easily accelerated with NRP discharges produced in the middle of the tube [91], while flame acceleration with NRP discharges close to the tube wall was more difficult to achieve [69]. Indeed, for discharges located in the boundary layer, only local ignition events and ignition kernel interactions with the leading shock upstream of the main flame could significantly accelerate the flame burning velocity. This result is interesting because it illustrates that the mechanisms of plasma action for flame acceleration can be quite complicated, especially in environments where plasma, flame, and acoustic/shock waves interact. These interactions will be further discussed in Section 4.

3.3. Lean stability and blow-off limits

Studies showing that non-equilibrium plasmas can extend the lean stability limit and/or the lean blow-off limit of a flame are among the most common in PAC. The general idea is to install a plasma source in a burner (typically swirl stabilized, bluff-body stabilized, or turbulent jet flames), or in the cavity of a scramjet, and to compare the lean stability and/or blow-off limits of the flame with and without plasma, for various bulk flows. Different plasma actuation strategies have proven to be successful. The three most popular are (a) with the discharges localized in the fresh gases [21], (b) with the discharges localized in the burned gases [64], and (c) with the discharges distributed in the fresh gases [67]. Various plasma sources have been utilized, for example NRP discharges [64], DBD [33], or gliding arcs [36,60,92]. Extension of the lean stability and blow-off limits have been observed for a large variety of conditions, including different fuels (methane, propane, kerosene, etc.), pres-

tures [25,93], and supersonic flows [60,80,81,94]. Except in some specific cases, noticeable effect of the plasma was observed for a plasma power of about 1% of the flame thermal power.

When the purpose of the plasma actuation is to extend the lean stability or blow-off limit, the repetition rate of the discharges has to be adapted to the flow rate (see Section 2.5). The residence time in the discharge area should be long enough for the gas to be properly “treated” by the plasma. For example, for discharges localized in the recirculation zone of a bluff-body [20,64], *i.e.*, in the burned gases, the velocity of the flow is usually low and the minimal repetition rate at which the discharges had a strong effect was 10 kHz. Higher repetition rates (up to 80 kHz), for a similar plasma power, did not improve further the plasma effect. For discharges localized or distributed in the fresh gases, this residence time matching is even more important. For example in Di Sabatino et al. [90], it was found that for high pulse repetition frequencies the plasma effect on the flame stabilization was the strongest. However, the flow disturbances produced by the discharges were also mentioned [90,95] and, similarly to what was obtained in plasma-assisted ignition studies, these could be detrimental to the flame stabilization.

In general, studies on the extension of the lean stability and blow-off limits by non-equilibrium plasma discharges mainly consider strong discharges, *i.e.*, discharges with a strong fast gas heating, that could be considered as local ignition sources [20,36,60,64,92,95,96]. With these discharges, it is easy to reach a ratio of plasma power to flame thermal power of about 1%, and therefore to obtain a strong effect on the flame. However, one of the strong limitations of these non-equilibrium discharges is the production of large quantities of nitric oxide, NO [33,67]. On the same setups, it would be possible to test weaker discharges, such as NRP glow discharges [53] or DBD, but if the plasma power is too low, the effect on combustion is marginal. However, recently a few studies showed a strong enhancement of the lean blow-off limit of methane-air swirl flames by discharges with a low thermal effect [33,93].

Di Sabatino and Lacoste [93] showed that at pressures higher than 2 bar, NRP glow discharges (with a low thermal effect) were more efficient than NRP spark discharges (with a strong thermal effect) in extending the lean stability and blow-off limits of methane-air swirl flames, as shown in Fig. 9. The ratio of plasma power to flame thermal power was kept constant for all pressures at 0.32% for NRP glow discharges (NRP Glows in Figs. 9 and 10a) and 0.7% for NRP spark discharges (NRP Sparks in Figs. 9 and 10a). For example, they showed that at 4 bar (Fig. 10a), the use of NRP glow discharges obtained with an applied voltage of 13 kV and a pulse repetition fre-

quency of 30 kHz, could save up to 650 W of thermal power, while maintaining a 16-kW flame. For the same conditions, NRP spark discharges could save 600 W only. The authors explained that the less efficient effect of the NRP spark discharges at elevated pressures could be due to the generation of strong shock waves close to the flame stabilization area [90]. However, they did not explain why NRP glow discharges became more efficient with increasing the pressure.

Based on the model developed by Castela et al. [82] (see Section 3.1), and the effect of pressure on the reduced electric field, an explanation can be proposed. For identical applied voltage and electrodes, an increase in pressure increases the gas density, N , and decreases the reduced electric field, E/N . As showed in Fig. 1, a decrease in E/N corresponds to a shift of the dissipated energy by electrons from electronic excitation to vibrational excitation of nitrogen. According to Castela's model, the energy of the NRP glow discharges can be split into two parts: electronic excitation of nitrogen, responsible for chemical dissociation and fast gas heating, and vibrational excitation of nitrogen, responsible for slow gas heating. Increasing the pressure will then favor the slow gas heating. At pressures above ambient, this slow heating due to vibrational-translational relaxation can happen in tens of microseconds [82]. With a discharge repetition rate of a few tens of kilohertz, this relaxation process can then be responsible for an almost constant gas heating in the interelectrode area. Thus, the main part of the discharge energy is deposited close to the flame stabilization area. On the other hand, at lower pressure, a larger fraction of the discharge energy goes into fast gas heating (higher E/N), generating gas expansion and transport effects that can be detrimental to the local plasma actuation, as observed in [83]. Temperature measurements as well as numerical simulations applied to this configuration would be valuable tools to validate this explanation.

In Kim et al. [33], the lean blow-off limit of a methane-air swirl flame was extended by using DBD generated by an AC power supply, with an applied voltage up to 7 kV and a frequency up to 4 kHz (see Fig. 10b). This study was performed at ambient pressure. For the entire range of bulk flow velocities studied (from 4 to 10 m/s), non-thermal plasma produced by DBD could extend the lean blow-off limit. This effect was attributed to a chemical effect with the production of ozone, O_3 , by the low temperature discharges. The plasma effect was compared with the effect of ozone addition, and they found that the DBD had a similar effect than an addition of 10,000 ppm of O_3 on the extension of the lean blow-off limit. However, a small thermal effect was not completely ruled out, as the concentration of NO in the burned gases increased by up to 20 ppm when the DBD were used.

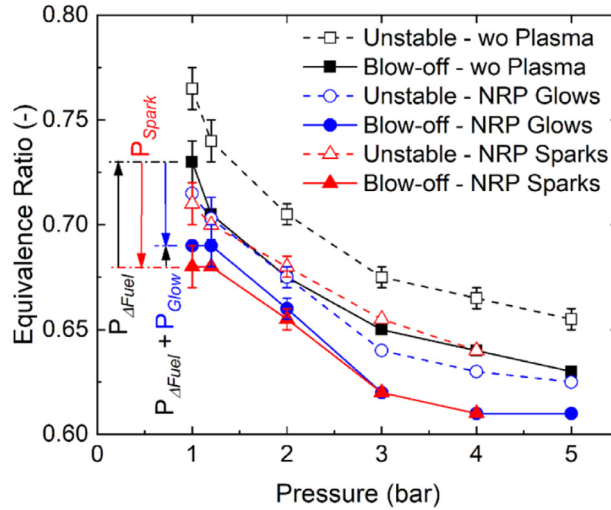


Fig. 9. Stability and lean blow-off limits of a methane-air swirl flame with and without NRP discharges as a function of pressure. Error bars represent standard deviation in reproducibility of results. Taken from [93].

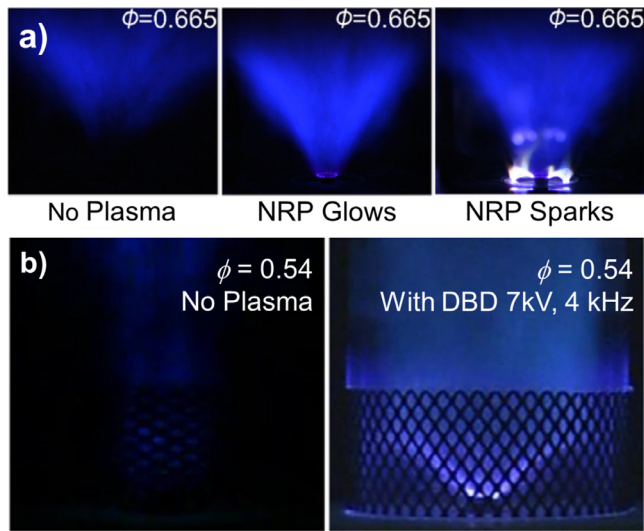


Fig. 10. a) Direct visualization of methane-air swirl flames at 4 bar, without plasma, with NRP glow discharges, and with NRP spark discharges. Adapted from [93]. b) Direct visualization of methane-air swirl flames at atmospheric pressure without plasma and with DBD. Adapted from [33].

Thus, the use of discharges favoring either a chemical effect or a slow gas heating might be the future of the lean blow-off limit extension by plasma. In order to have the maximal effect on the flame, with a low plasma power and low NO emissions, additional efforts on the optimization of the plasma location and an increase of the plasma volume and the repetition rate of the discharges are necessary. These parameters are conditions dependant, therefore, ideally, these studies should be done in configurations close to the targeted applications.

3.4. Modeling of flames with plasmas

Modeling and simulations of PAC are less common than experimental studies but they have played a major role for the understanding of the behavior of flames with plasmas. These studies can be grouped into two subsets (1) the development of chemical kinetic models, and (2) the development of computational fluid dynamics (CFD) codes including plasma models.

There are two challenges in the development of chemical kinetic models for plasma-assisted com-

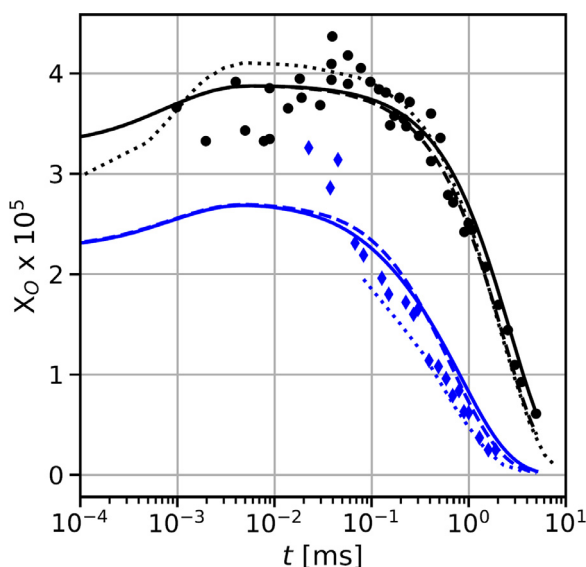


Fig. 11. Comparisons of the temporal evolution of the atomic oxygen mole fraction in air (black) and methane-air (blue) following a 100-ns discharge. Experimental data (symbols) from [104] and simulations from [104] (dotted lines) and from [103] (solid and dashed lines). Taken from [103].

bustion problems. First, the plasma temporal scale is at least three orders or magnitude smaller than the combustion temporal scale. Second, while the reaction rates of combustion are Arrhenius' functions of the gas temperature, plasma reaction rates are usually functions of the reduced electric-field. A common way to overcome these challenges is to consider that the discharge and the combustion process happen in a sequential way [97]. During the discharge phase, only the plasma chemistry is considered. At the end of the discharge, the plasma composition and the thermodynamic conditions are then used as the initial conditions for the combustion simulation. Other coupling strategies can also be found, such as a frozen electric-field approach in [98], or a 0-D calculation for the discharge followed by a 1-D calculation for the combustion [99].

While the chemistry of air-plasmas is relatively well known [100,101], the development of chemical kinetic models for plasmas of gas mixtures relevant for combustion environments is still ongoing [51,102,103]. One of the difficulties in the development of these models is to include only the species that are essential for the plasma and the combustion chemistry, in order to keep the size of the models compatible with multi-dimensional calculations. The identification of these species is done by combining a modeling and an experimental approach [104–110]. For example, Fig. 11 presents the comparison of the evolution of atomic O mole fraction following a 100-ns discharge in air (black) and in a methane-air mixture (blue) at low pressure [104].

The 0-D simulations were performed with a plasma model including 71 species and 680 reactions [103].

Note that, in some PAC configurations (see for example [27,64]), the discharges are generated in the burned gases. Chemical kinetic models of plasmas of nitrogen, oxygen, water, and carbon dioxide mixtures have yet to be developed and validated.

Multi-dimensional CFD calculations of PAC problems usually do not include detailed plasma chemistry. Reduced models with a minimal number of charged and excited particles [97,98,111–113] or phenomenological plasma models [82,83] are often preferred. The vast majority of these numerical studies were 1-D or 2-D simulations, focusing on ignition problems for which the localisation of the discharge was *a priori* known, sometimes including multiple discharges, representative of NRP discharges. To the best of my knowledge, CFD simulations of flame stabilization by distributed discharges such as in [9,33,93,114] have not been achieved, yet. Further work in plasma modeling and in flame-plasma coupling is still required to allow these multi-physics simulations.

4. Application to flame dynamics

In combustion systems where a flame is continuously burning, problems of flame motion or fluctuation can arise. Flame dynamics refer here to these undesired or uncontrolled changes in the flame stabilization area, surface or shape. Different instability mechanisms can be responsible for

these changes. One of the most studied is the coupling with acoustic waves, also known as thermoacoustic coupling or thermoacoustic instabilities. In gas turbine engines, problem of thermoacoustic instabilities is of utmost importance. The susceptibility of flames to upstream perturbations and their coupling with resonant modes of the system has been one of the biggest challenges in the development of modern high-efficiency, low-emission gas turbine engines in recent decades [16]. The occurrence of flame instabilities is generally considered as one of the highest risk items in new engine development [115]. While analysis and simulation of thermoacoustic stability has made great progress in recent years [116], fully predictive modeling is still challenging.

In ground-based gas turbines, thermoacoustic instabilities can be hindered by passive control systems. Passive systems use clever geometrical arrangements of the combustion chamber elements to avoid thermoacoustic coupling. They most often rely on the use of well-tuned resonators to damp specific dangerous oscillation modes of the system, which have been identified during the design and prototyping phases. Passive techniques however have severe drawbacks: they generate large additional expenses during the engine development and they lead to an additional weight of the engine, which severely penalizes these solutions for aero-engine gas turbines. For power generation applications, the cooling and purge mass flow required for the damping resonators has an adverse effect on the system efficiency. Therefore, alternative solutions such as active control strategies are desirable. Their concept is to alter the flame dynamics or the system's acoustics by actively detuning the coherent response between the flame and pressure oscillations using sensors and actuators.

Non-equilibrium plasma actuators have shown promising potential for active control of thermoacoustic instabilities. Recent experimental studies have demonstrated the capability of non-equilibrium plasma discharges to control flame dynamics [20–22,25,67,95,114,117–121], without the drawbacks of traditional actuators such as loudspeakers or fuel valves [122]. Two groups of studies can be differentiated: studies on the effect of non-equilibrium plasma discharges on the flame response to acoustic perturbations, and studies demonstrating a mitigation of thermoacoustic coupling. The key results are presented in the next sections, followed by a discussion about what is missing before considering non-equilibrium plasma discharges as an actuator to actively control flame dynamics in gas turbine engines.

4.1. Flame transfer functions

To stop a thermoacoustic instability, the two main options are to alter the flame response to the perturbations or to modify the acoustic field of the

system. Except for a few studies reporting on the use of non-equilibrium plasma discharges as an acoustic source [123], scientists have been mainly considering non-equilibrium plasmas to modify the flame response. The flame response to acoustic perturbations depends on the amplitude and frequency of the acoustic waves. For perturbations of small amplitude, the flame response is linear and only depends on the frequency. Thus, it can be studied in a systematic manner by using the flame transfer function (FTF) formalism [124].

When a flame is subjected to velocity perturbations of the flow, u' , the heat release rate fluctuates by an amount $\dot{Q}'$. The FTF is defined as the ratio of the relative heat release rate fluctuations and the relative velocity fluctuations, in the frequency domain:

$$F(\omega) = \frac{\hat{\dot{Q}}(\omega)/\bar{\dot{Q}}}{\hat{u}(\omega)/\bar{u}}, \quad (4)$$

where ω is the frequency, $\dot{Q}$ is the heat release rate, u is the flow velocity at the flame location, $\hat{X}$ is the Fourier transform of the quantity X , and $\bar{X}$ is the mean value of X . It can also be expressed in terms of a gain, G , and a phase, ϕ :

$$F(\omega) = G(\omega)e^{i\phi(\omega)}, \quad (5)$$

where G reflects the magnitude of the flame response and ϕ defines the time lag between the velocity and the heat release rate fluctuations, at the base of the flame.

To measure the velocity at the flame base, optical techniques such as laser Doppler velocimetry (LDV) can be used. However, other techniques such as multi-microphone method [125] or hot-wire anemometry may be preferred. Measurements of heat release rates can be more challenging. However, in the case of perfectly premixed flames, it has been shown that fluctuations of heat release rate are proportional to fluctuations of chemiluminescence of either the excited methyl radical, CH^* , or the hydroxyl radical, OH^* [126,127]. Thus, a camera or a photomultiplier tube (PMT), equipped with an adequate band pass filter can be utilized.

In 2013, two papers [114,117] demonstrated that non-equilibrium plasma discharges could significantly modify the response of methane-air flames to acoustic perturbations. Volkov et al. [117], showed that a DC corona discharge could increase the FTF gain of a laminar flat flame (stabilized over a porous plate) for frequencies higher than 150 Hz. They attributed this result to a transport effect by ionic wind. A few months later, Lacoste et al. [114] demonstrated that NRP spark discharges could significantly decrease the FTF gain of a 4-kW swirl flame, for a range of frequencies from 8 to 350 Hz. They did not provide an explanation for this result but they mentioned that it could be attributed to a thermal effect, a chemical effect, or a hydrodynamic effect, due to gas expansion and

acoustic waves generation following fast gas heating. These two first papers are interesting because they show that different plasma actuation strategies can significantly change the flame dynamics. The next step was then to identify which plasma is the most efficient.

To compare different plasma strategies, it is important to select non-equilibrium plasma sources promoting different effects. In 2017, Lacoste et al. [9] compared the response of a laminar methane-air flame to acoustic forcing, forcing by an AC electric field, and forcing by NRP glow discharges. The AC electric field was chosen because its main effect was to generate ionic wind from the flame, while NRP glow discharges were selected due to their minimal thermal and transport effects, therefore promoting a chemical actuation of the gas. New definitions of FTF for forcing by electric fields, $F_E(\omega)$, and plasma, $F_P(\omega)$, were proposed:

$$F_E(\omega) = \frac{\hat{Q}(\omega)/\bar{Q}}{\hat{V}(\omega)/\bar{V}} = G_E(\omega)e^{i\phi_E(\omega)}, \quad (6)$$

where V is the applied voltage and $\bar{V}$ is a reference voltage (1 V). Here also, the FTF can be expressed in terms of a gain G_E and a phase ϕ_E . Note that the denominator of this transfer function being independent of the average flame thermal power, this definition does not allow quantitative comparisons between gains obtained for forcing by acoustic waves and electric field. For plasma forcing:

$$F_P(\omega) = \frac{\hat{Q}(\omega)/\bar{Q}}{\text{sinc}(1/2)P_P(\omega)/\bar{Q}} = G_P(\omega)e^{i\phi_P(\omega)}, \quad (7)$$

where P_P is the average plasma power deposited during half a period of the forcing. The term $\text{sinc}(1/2)$ comes from the Fourier expansion of the square-wave plasma forcing (more details in [9]). The terms G_P and ϕ_P are the gain and the phase of the FTF, respectively. The amplitude of the forcing is defined as $P_P/\bar{Q}$, thus, quantitative comparisons between the FTF gains for forcing by acoustic waves and NRP glow discharges are possible.

The authors showed that, qualitatively, the flame was similarly responsive to acoustic, AC electric field, and plasma forcing, for the entire range of frequencies studied, as presented in Fig. 12. They explained this result by the fact that electric field forcing was actually inducing a change in the local flow velocity, which is similar to velocity fluctuations induced by acoustic waves. When NRP glow discharges were applied, the flow field was not affected but the local reactivity of the gas was increased close to the tip of the flame. This change in the local reactivity could be modeled as a local increase in the flame burning velocity, having a similar kinematic impact that a change in the local flow velocity [88].

These results were obtained for a laminar flame of about 100 W. Could they be extrapolated to turbulent high-pressure flames? Recently, Deng et al. [120] investigated the effect of DBD produced by an AC voltage source on the transfer function of a turbulent swirl flame at atmospheric pressure. They showed that below the breakdown voltage (4 kV in their case), for which only an ionic wind effect can be expected, the FTF was not affected by the electric field. This result could have been expected because the ion density in a flame is too low to generate strong ionic winds. If the ion density can be increased, for example by using DBD in the flame or in the flow upstream of the flame, then the flow generated by ionic wind might be strong enough to modify the turbulent flame response to acoustic perturbations. When they increased the applied voltage and DBD were produced, with a coupled transport and chemical effect (see Section 2.1), the FTF gain was significantly decreased. However, with their configuration, it is not possible to distinguish if this result was obtained mainly due to a chemical effect, or to a coupled chemical-ionic wind effect. Regarding the effect of non-equilibrium plasma discharges on the transfer functions of flames at elevated pressures, to the best of my knowledge, there is no published results yet.

Thus, while it is established that non-equilibrium plasma can mitigate the response of swirl turbulent flames to acoustic perturbations, and demonstration has been made that chemical actuation and ionic wind could have a strong effect on the flame dynamics, the question of the best plasma actuation for practical applications remains open.

4.2. Control of thermoacoustic instabilities

A few papers report on the active control of thermoacoustic instabilities by non-equilibrium plasma discharges [19,21,22,25,114,121]. In all cases, the thermoacoustic coupling happened between longitudinal acoustic waves and premixed flames, and the non-equilibrium plasmas were produced by NRP discharges. Two types of control strategies have been attempted: by continuously applying the non-equilibrium plasma discharges until the flame stabilizes, or by forcing the flame with the plasma, with an appropriate phasing with respect to the acoustic fluctuation, *i.e.*, with a feedback loop.

In 2013, Moeck et al. [19] showed that for a 43-kW swirl flame of natural gas and air at atmospheric pressure, a thermoacoustic instability at 150 Hz could be completely suppressed by applying NRP spark discharges at 50 kHz, continuously. For these conditions, the plasma power was about 315 W, *i.e.*, about 0.7% of the flame thermal power. Later on, these results were confirmed by similar results [21,22,114]. Recently, Kim and Cohen [25] extended these results to a pressure of 5 bar. This

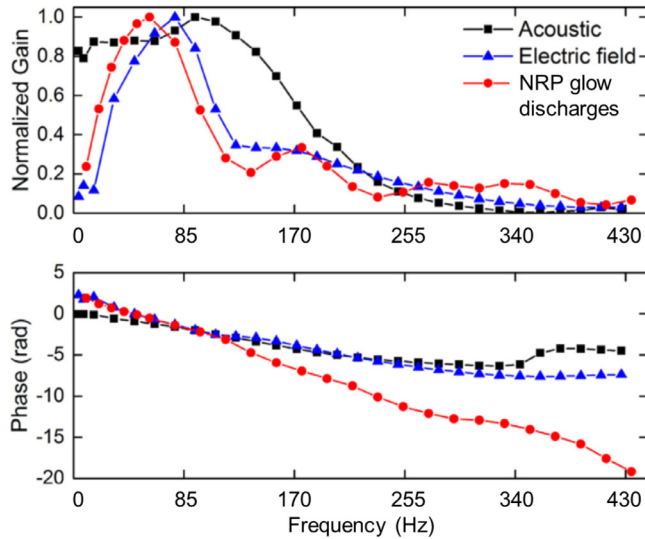


Fig. 12. Comparison of the transfer functions of a methane-air V-flame to forcing by acoustic waves, AC electric field, and NRP glow discharges. Adapted from [9].

pressure was chosen because it is representative of micro-gas turbine engines, or of a typical low-power regime of aero-engine combustors. The authors noticed that to obtain a similar plasma effect when the bulk flow velocity and/or the pressure were increased, the electric power deposited in the NRP spark discharges should be linearly increased.

To decrease the overall plasma power deposited, *i.e.*, to increase the plasma actuation efficiency, instead of applying continuously the discharges, they can be gated at a chosen frequency. The idea is that NRP discharges produce a forcing that counteract the acoustic perturbation. For example, for a thermoacoustic instability at 120 Hz, *i.e.*, with a period of 8.3 ms, NRP discharges at 30 kHz can be applied during 4.15 ms, which corresponds to 126 pulses, then no discharges are applied during 4.15 ms, before applying again 126 pulses at 30 kHz, and so on. This is what was done in [19], for an atmospheric pressure natural gas and air swirl flame of 41 kW. The authors showed that, depending on the phasing between the acoustic oscillations and the plasma forcing, the amplitude of the peak pressure induced by the thermoacoustic coupling could be either decreased or increased. This was the first evidence that non-equilibrium plasma discharges could be used in a closed-loop control scheme to mitigate thermoacoustic instabilities. Again, in this study, as the spark regime of NRP discharges was utilized, the authors could not conclude on the main mechanism of plasma action.

Recently, a similar approach was used with NRP glow discharges [121]. Recall that for the glow regime of NRP discharges, the fast gas heating is minimal, there is no transport effect, and the chemical actuation of the gas is promoted (see

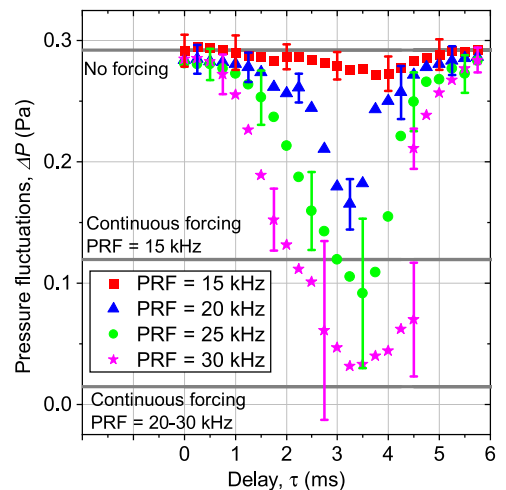


Fig. 13. Amplitude of the pressure fluctuations, ΔP , for different pulse repetition frequencies, PRF, as a function of the delay, τ between the plasma forcing and the measured acoustic wave. The NRP glow discharges were obtained for an applied voltage of 6.8 kV. The duty cycle of the forcing was 50%, and the frequency of both the forcing and the acoustic wave was 170 Hz. Taken from [121].

Section 2.4). Fig. 13 presents the pressure fluctuations of a 230-W wall stabilized laminar flame of methane and air, without and with plasma forcing. Without plasma forcing, the flame experiences a thermoacoustic instability at 170 Hz, with an amplitude of the pressure fluctuations, ΔP , of 0.3 Pa. For NRP glow discharges continuously applied at 30 kHz, the pressure fluctuations could be com-

pletely suppressed, but the plasma power was about 1.6% of the flame thermal power. Applied in a closed-loop control scheme with a duty cycle of 50%, the thermoacoustic instability could be suppressed for a plasma power of 0.8%. Even if this was obtained for a tiny flame, this result is promising for applications at elevated pressures. Indeed, as discussed for lean flame stabilization in [Section 3.3](#), NRP glow discharges seems to be more efficient at elevated pressure than NRP spark discharges.

4.3. Towards engine conditions

Applications of plasma-assisted flame dynamics mainly target combustion problems in gas turbine engines. Thus, even though non-equilibrium plasma demonstrated good potentials for flame stabilization, the conditions at which the results were obtained are so far from real engine conditions that it is not possible to conclude about the relevance of a plasma actuator. Over this decade, we learned that at pressures up to 5 bar, NRP discharges can stabilize turbulent methane-air or propane-air flames of a few tens of kilowatts, by applying a plasma power of about 1% of the flame thermal power. We learned that separately, different effects of the plasma can be efficient in flame stabilization but we do not know which effect, or which combination of effects, is the most efficient for flame stabilization. To gain confidence that a plasma actuator can work in a real gas turbine engine, further work is needed.

First, the question of the efficiency of plasma actuation on other modes of thermoacoustic instabilities is still open. As highlighted by Poinso in [\[116\]](#), the vast majority of studies on flame dynamics focus on longitudinal acoustic waves, while in practice transverse acoustic waves or azimuthal acoustic waves can be as, or even more, detrimental. These other acoustic modes can be significantly different than the longitudinal one, with, for example, standing or spinning waves, or featuring much higher frequencies. How the discharges could affect the flame-acoustic coupling in these conditions is unknown, and the plasma-acoustic coupling is unknown as well. This later coupling for non-thermal discharges has been barely studied in atmospheric pressure air [\[123,128\]](#), but the results showed strong interactions between the acoustic field and the plasma discharges. In addition, aside from acoustic perturbations, turbulent flames are also responsive to a broad range of disturbances, *e.g.*, flow, composition, temperature, or enthalpy. Recently, Sun et al. [\[26\]](#) showed that gliding arc discharges could enhance the lean blow-off limit of methane-air swirl flames perturbed by low frequency flow velocity steps but further work is required to demonstrate plasma actuation for more complex or coupled disturbances.

Another important parameter is the gas density.

Depending on the use (propulsion or power generation), the combustor pressure and temperature in gas turbine engines can be very different, typically from 3 to 45 bar and from 500 to 800 K [\[129\]](#). The gas density is crucial for the properties of electrical discharges. In addition to the obvious effect on the reduced electric field controlling the plasma chemistry, the gas density also affects the dimensions of the discharges. At low pressures, discharges are usually more diffuse than at higher pressures but the pressure effect is highly non linear [\[4\]](#). For example, Xu et al. [\[75\]](#) showed that in air, the initial radius of NRP spark discharges of 6.2-mJ energy per pulse decreased from 220 to 150 μm when the pressure was increased from 1 to 5 bar, while between 10 and 20 bar, the radius remained almost constant at about 120 μm . This decrease in the discharge volume with pressure can significantly increase the strength of the shock waves from the discharge area (as discussed in [\[90\]](#)). Therefore, the main mechanism of plasma effect could transition from thermal to transport. Similarly, while the glow regime of NRP discharges was disregarded for PAC at atmospheric pressure, mainly because (1) the spark regime is very efficient and (2) it is difficult to obtain enough plasma power without transitioning to the spark regime, this regime is really promising at elevated pressure.

The question of pollutant emissions in non-equilibrium plasma discharges will also have to be taken into account during the design of industrial plasma actuators. In fuel-air mixtures, similarly to what is happening in combustion, the production of NO by the discharges is significant when the gas temperature is high (typically higher than 1800 K [\[1,105\]](#)). This local production of NO in the discharges can be large enough to drastically increase the global NO concentration in the burned gases of the combustion system [\[19,25,130–132\]](#). For example, in [\[19\]](#), NRP spark discharges at 20 kHz could mitigate the thermoacoustic instability of a 43 kW flame, but the concentration of NO in the exhaust gases was increased from 2 to 17 ppm [\[67\]](#). Thus, focusing on optimizing the impact on flames of discharges with low gas heating, such as DBD or NRP glow discharges, could be a good strategy to avoid further issues of pollutant emissions.

Finally, while gas turbine engines are burning a large variety of fuels, from low to high carbon content such as methane, syngas, kerosene, heavy fuel oil, or coal, PAC studies using non-equilibrium plasma discharges have mainly utilized methane, propane and ethylene, with only a few studies dealing with liquid fuels (n-heptane [\[27,133\]](#), dodecane [\[27\]](#), iso-octane [\[133\]](#), or kerosene [\[134\]](#)). The interactions between plasma discharges and liquid droplets can change the effect of plasma on the flame dynamics. In addition, as mentioned in [\[25\]](#), the plasma effect is fuel dependant with, for a same plasma power, a stronger effect on methane flames

than on propane flames. Note that plasma-assisted coal combustion has been intensively studied but mainly with thermal plasma (see for example [135–137]). It would be very interesting to have more information about plasma-liquid fuel interactions as well as on the chemistry of plasmas of mixtures of large hydrocarbons and air. Finally, with the recent shift of research interest towards carbon-free fuels, recent papers on plasma-assisted ammonia flames have been published [138–140]. This research is much needed for an efficient transition to zero-carbon emissions.

5. Conclusion

This paper first reviews some of the most commonly used non-equilibrium plasma sources for plasma-assisted combustion (PAC) studies: DBD, gliding arcs, microwaves, and NRP discharges. Then a review of the results obtained for ignition, acceleration of the burning velocity, and enhancement of the lean blow-off limits is proposed, followed by a more detailed discussion on the utilization of non-equilibrium plasma for the control of flame dynamics, targeting application in gas turbine engines. Main conclusions are:

- While ignition by non-thermal plasma discharges have been intensively studied in CVCC and in continuous flow burners, there is a lack of experimental data validating chemical models of PAC and under conditions relevant to engines, *i.e.*, with commercial fuels and at thermodynamic and flow conditions far from ambient.
- Fast gas heating obtained for discharges in air containing mixtures, for reduced electric fields in the range 100 to 1000 Td, can be favorable or detrimental to combustion processes. It induces gas expansion and generates shock/acoustic waves, responsible for local flow disturbances and removal of a part of the discharge energy from the plasma location.
- The efficiency of plasma action is strongly setup dependant. As the discharges usually have a volume significantly smaller than the flame volume, their location is crucial for the combustion enhancement. In addition, increasing their volume is highly desirable and strategies promoting large volumes of non-equilibrium plasmas in combustion environments should be further developed.
- Flame dynamics can be controlled by non-equilibrium plasma discharges obtained from various sources. However, which energy pathway (thermal, chemical, or transport) should be promoted remains unclear. More studies comparing each of these effects, separately and combined are necessary to design an efficient plasma actuator.

- While the effects of pressure, mixture composition, and flow are significant on combustion processes, their effects are as important to the physics of plasma and relatively poorly documented for conditions relevant to combustion applications. Fundamentals studies of plasma at pressures higher than atmospheric as well as in different flows and gas mixtures are still lacking. So are PAC studies in engine relevant conditions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

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